

Hurricane Relief App

Software Requirements Specification



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Basement Coders

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1. Introduction

Purpose

The Hurricane Relief App is designed to provide first responders and civilians with accurate, easy-to-understand hurricane information before, during, and after a storm. It will deliver real-time warnings, local hazard updates, evacuation information, and safety guidance in one simple interface.

For first responders, the app will support faster decisions by showing storm impacts, dangerous areas, road conditions, shelters, and critical updates. For civilians, it will help them understand the threat to their location and clearly explain what actions they should take to stay safe.

The goal is to turn reliable hurricane data into simple, timely information that helps communities prepare, respond, and recover.

2. Stakeholders



ALEX MORGAN REYES

HURRICANE SPECIALIST



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Gulf Coast Region, USA

BIO

Dedicated and experienced hurricane response specialist with a passion for protecting communities and coordinating rapid disaster relief efforts in high-pressure environments.

SKILLS

- Incident Command System (ICS)
- Search & Rescue Operations
- Evacuation Planning
- Disaster Risk Assessment
- Community Outreach
- First Aid & CPR Certified
- Logistics & Supply Chain
- Crisis Communication
- Multi-Agency Coordination

EDUCATION

Bachelor of Science in Emergency Management – Louisiana State University

Completed a comprehensive program focused on disaster preparedness, crisis leadership, and large-scale incident coordination. Developed a strong foundation in hazard mitigation, emergency operations planning, and interagency communication, with specialized coursework in coastal storm impacts and critical infrastructure resilience. Participated in LSU's Gulf Coast disaster simulation labs, gaining hands-on experience in hurricane response modeling, evacuation route analysis, and resource allocation strategies for high-risk communities. Engaged in applied research on multi-agency coordination during Category 3–5 storms, strengthening operational decision-making skills essential for real-world disaster environments.

EXPERIENCE

LEAD HURRICANE RESPONDER – FEMA REGION VI

June 2018 – Present

Coordinate multi-agency hurricane response operations to ensure rapid, unified deployment of personnel, equipment, and supplies to impacted regions. Lead comprehensive evacuation planning and shelter-in-place strategies for coastal communities, emphasizing safety, communication, and resource efficiency during Category 3–5 storms.

PAIN POINTS

- Disjointed multi-agency coordination Emergency, state, federal, and nonprofit partners often operate on separate timelines and priorities, creating delays and confusion during critical response windows.
- Slow deployment of life-saving resources Bottlenecks in logistics, transportation, and communication can prevent supplies, personnel, and equipment from reaching impacted areas quickly
- Ineffective or outdated evacuation strategies Many coastal communities lack clear, scalable plans for Category 3–5 storms, leading to last-minute evacuations, unsafe routes, and overwhelmed shelters.



LIV PSAROUDIS

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SKILLS

- Species identification and taxonomic classification
- Water quality analysis (salinity, pH, dissolved oxygen, turbidity)
- SCUBA diving (certification level can be specified — e.g., AAUS Scientific Diver)
- Underwater photography and videography for survey documentation
- GIS mapping and spatial data analysis
- Statistical analysis software (R, SPSS, or Python for ecological data)
- Tissue sampling and necropsy procedures
- Tagging and tracking technology (acoustic telemetry, satellite tags)

NEEDS

- Accurate storm surge and coastal flooding predictions to protect research stations and equipment near shorelines
- Advance warning windows (72+ hours) to secure or relocate field equipment, boats, and monitoring buoys
- Post-storm damage assessment tools to plan recovery surveys quickly

EXPERIENCE

Marine Biology Research Assistant — Oceanic Conservation Institute (2022–2024)

- Conducted coral reef health surveys across 15+ dive sites, tracking bleaching rates and biodiversity
- Collected and analyzed water quality data to support NOAA compliance reporting
- Co-authored a peer-reviewed paper on reef fish population decline

Field Technician — Gulf Coast Marine Sanctuary (2020–2022)

- Assisted with sea turtle tagging and migration tracking using satellite telemetry
- Maintained field databases for long-term population monitoring
- Led public outreach tours educating visitors on local marine ecosystems

PAIN POINTS

- No marine-specific data — existing apps focus on coastal population centers and property, not ocean conditions relevant to her research (water temp shifts, wave height at reef sites, current changes)
- Poor offline functionality — field sites often have unreliable signal, and apps like Clime or Tropical Hurricane Tracker aren't built for use without connectivity
- Generic alert zones — push alerts are based on cities/counties, not marine sanctuaries, dive sites, or specific research coordinates she needs to track



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GOALS

- See current hurricane conditions, flood zones, road closures, and evacuation routes in one place.
- Quickly identify high-risk areas, shelters, hospitals, fire stations, and residents needing assistance.
- Assign crews and track their status during response operations.
- Receive reliable alerts even when internet or cellular service is limited.
- Document incidents, rescues, and damage for shift handoffs and after-action reports.

NEEDS

A mobile-first, easy-to-read dashboard with offline access that combines weather alerts, live maps, road and flooding reports, crew locations, incident assignments, shelter capacity, and emergency checklists. It should prioritize the most urgent information and let Mark send quick status updates to his team.

EXPERIENCE

LIEUTENANT / EMERGENCY RESPONSE
COORDINATOR AT A COASTAL FIRE DEPARTMENT
June 2012 – Present

Mark has spent 14 years as a firefighter along the Gulf Coast. During hurricane season, he helps coordinate evacuations, rescues, station readiness, and post-storm damage response. He works long shifts with limited cell service, changing road conditions, flooded neighborhoods, and frequent power outages.

His department currently relies on radio updates, weather websites, group texts, and paper maps. Information is often scattered or outdated, making it difficult to decide where crews should go first.

PAIN POINTS

- Weather and emergency information changes faster than his team can communicate it.
- Flooded or blocked roads can turn a short response into a dangerous delay.
- Dispatch, police, EMS, utility crews, and shelters may each have different information.
- During major storms, exhausted crews need simple tools that work quickly under pressure.
- He cannot depend on a complicated system, especially while wearing gloves or working in rain.

3. Constraints

1. Time Constraints

- The project must be fully completed within a **3-month development window**, including planning, implementation, testing, and documentation.

2. Monetary Constraints

- The project has been allocated a **budget of \$0**, requiring all development to rely exclusively on free tools, open-source resources, and existing infrastructure.

3. Technical Constraints

- All source code must be written in **Java**, adhering to standard Java development practices and compatible with commonly available Java runtimes.
- No paid libraries, frameworks, or external services may be used due to the zero-budget requirement.

4. Overall Description

The Hurricane Tracker is designed to provide precise, real-time storm monitoring through a clean, intuitive interface. Unlike competitors that overwhelm users with broad, cluttered weather dashboards, our platform is purpose-built for hurricane-specific intelligence.

Our system focuses on delivering live updates, clear visualizations, and easy-to-understand data, ensuring that users receive reliable information quickly—especially during critical moments. By

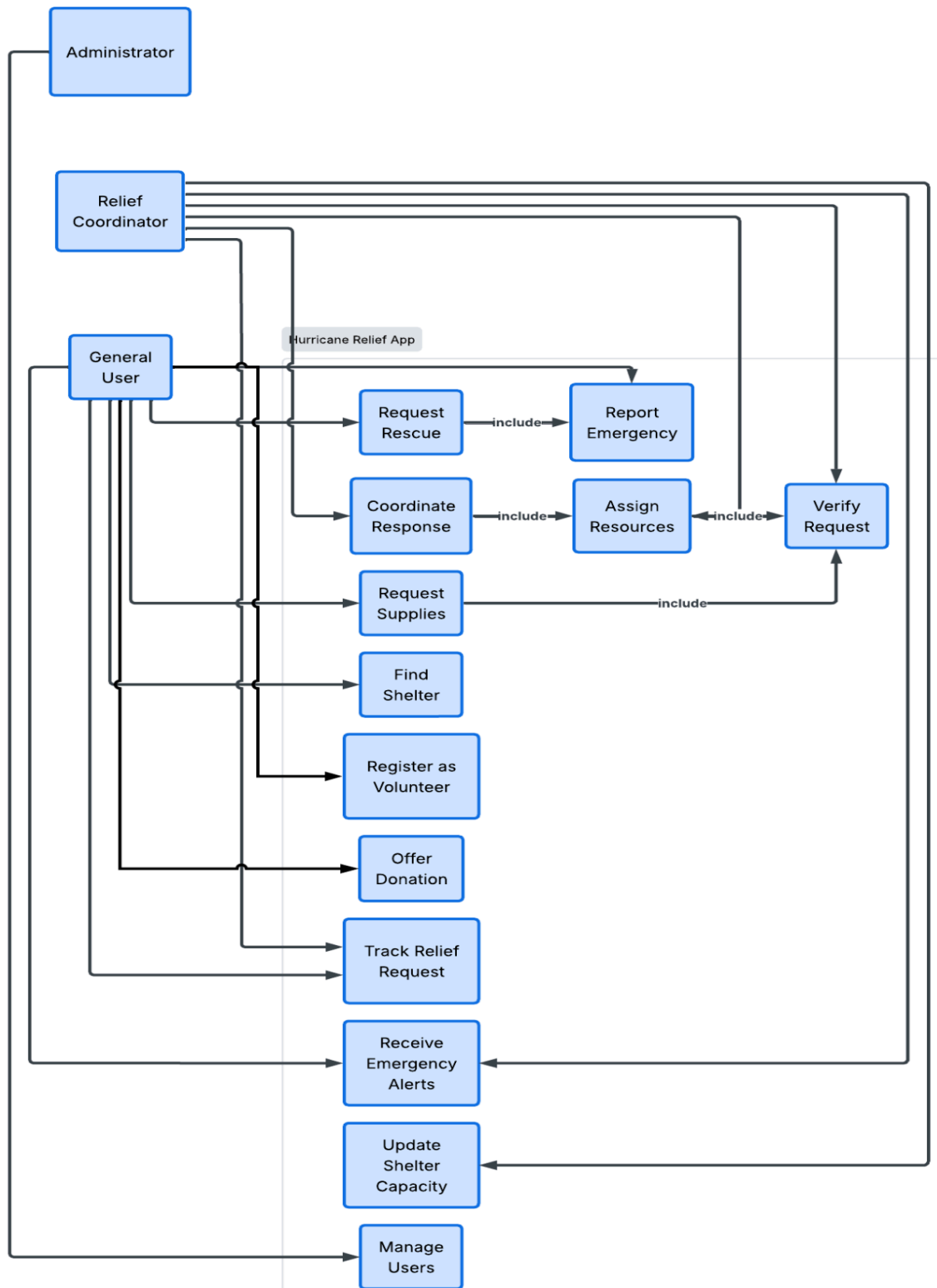
prioritizing accuracy, simplicity, and speed, the Hurricane Tracker empowers individuals, families, and communities to make informed decisions before, during, and after major storm events.

Key strengths include:

- Real-time hurricane tracking with minimal latency
- Streamlined interface designed for clarity under pressure
- Focused feature set that avoids unnecessary noise
- Reliable data sourcing for trustworthy storm insights

This product is built for users who need fast, dependable, and straightforward hurricane information without the distractions of full-scale weather platforms.

5. Business Use Cases



6. Functional Requirements

7. Non-Functional Requirements

8. Definitions and Acronyms

Definitions

System — The hurricane relief application being described in this document.

User — Any person who accesses the system, including hurricane victims, relief coordinators, and administrators.

Shelter — A designated location where affected individuals can seek safety during or after a hurricane.

Relief Coordinator — A staff or volunteer role responsible for managing resources, requests, and updates within the system.

Uptime — The amount of time the system is running and available for use, usually shown as a percentage.

Encryption — A method of converting data into a coded form so it cannot be read without the correct key.

Role-Based Access Control — A security method that limits what a user can see or do based on their assigned role.

Failover — A backup process that automatically switches to a working system if the main one fails.

Latency — The delay between a user's action and the system's response.

Degraded Mode — A limited version of the system that still works under poor network conditions, such as text-only alerts.

Acronyms

NFR — Non-Functional Requirement

UI — User Interface

UX — User Experience

API — Application Programming Interface

TLS — Transport Layer Security

WCAG — Web Content Accessibility Guidelines

FEMA — Federal Emergency Management Agency

NWS — National Weather Service

MTBF — Mean Time Between Failures

CSV — Comma-Separated Values
JSON — JavaScript Object Notation

9. Competitive Analysis

Competitor Name	Services	Strengths	Weaknesses
AccuWeather Hurricane Tracker http://www.accuweather.com/en/hurricane	Tracks active and past hurricanes; provides storm updates, news, preparedness resources, radar, and mobile alerts.	Easy to use; plain-language storm information; good for public awareness; shows global storm activity.	Built for consumers, not first responders, resource management, local incident maps, and operational checklists.
Clime https://climeradar.com/weather-radar/it-rome-41485	Uses real time radar imagery and detailed weather maps to give current updates to users.	Official NOAA affiliation, broad weather coverage in premium plan with many different weather events covered.	Pay for premium features; essentially a weather app without a subscription.
Tropical Hurricane Tracker (JPL Tech) https://apps.apple.com/us/app/tropical-hurricane-tracker/id1436318547	Covers various hurricanes, weather systems, and disturbances across oceans.	Easy to use interface and active response team to respond to feature requests and bug reports.	Built by a small tech team with low social proof and actual user count. Marketed to be a “less cluttered” app but no tech advantages.

10. References

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Federal Emergency Management Agency (FEMA). *National Response Framework*. U.S. Department of Homeland Security.

National Weather Service (NWS). *NWS API Documentation*. National Oceanic and Atmospheric Administration. <https://www.weather.gov/documentation/services-web-api>

Internet Engineering Task Force (IETF). *RFC 8446: The Transport Layer Security (TLS) Protocol Version 1.3*.

11. Appendices

Appendix A: Assumptions and Constraints

The system assumes users have access to a smartphone, tablet, or computer with an internet connection during or after a hurricane. The system assumes power and internet outages may occur, which is why a degraded mode is required. The system is constrained by the availability of third-party data, such as National Weather Service and FEMA feeds, and cannot guarantee accuracy beyond what those sources provide.

Appendix B: User Roles

General User — A person affected by a hurricane who needs information, shelter locations, or to request help.

Relief Coordinator — A staff or volunteer who manages requests, updates shelter status, and coordinates resources.

Administrator — A user with full system access who manages accounts, permissions, and system settings.

Appendix C: Sample Use Case

Use Case: Find Nearest Shelter

Actor: General User

Description: A user affected by a hurricane opens the app and searches for the nearest open shelter. The system displays a list of shelters sorted by distance, along with capacity and status.